

Microbes in Water



Drinking water: Water that is safe to consume and free of pathogenic organisms and chemical substances that are harmful to human health.

Obtained from

Rain water

Surface water (rivers and streams)

Stored water

Ground water(wells and springs- filters
through successive layers of sand).

Sea water

Quality standards of drinking water

Physical: colour, clarity, taste, odour, amount of suspended solids

Chemical: pH, amount of dissolved solids, hardness, amount of nitrogenous matter

Biological: Bacteriological content, amount of dissolved oxygen (BOD)

Properties required for drinking water

Tasteless, colourless, cool, clear and free of pathogens.

Drinking water guidelines which are followed:

US public health service

World health organization (WHO)

Environmental protection agency (EPA).

Water pollution

- Any physical or chemical change in water that adversely affects the health of the human and other organisms.

Water Pollution and Eutrophication

Pathogens

Organic wastes

Sediment

Chemical pollutants

Nutrients

Eutrophication: an enrichment problem

- Lake, estuaries and slow flowing streams that have minimal levels of nutrients are unenriched called as an oligotrophic lake.
- It has clear water and small populations of aquatic organisms.

Eutrophication is the enrichment of a lake, estuary or slow flowing stream by inorganic plant and algal nutrients.

The enrichment of water results in an increased photosynthetic productivity. The water in a eutrophic lake is cloudy and usually has vast numbers of algae and cyanobacteria.



Pollution ✉ effects man, flora and fauna

Addition of chemicals to water

Pesticide residues

Other toxic chemicals

Algal toxins

Fecal contamination (microbial) ✉ Water borne
diseases

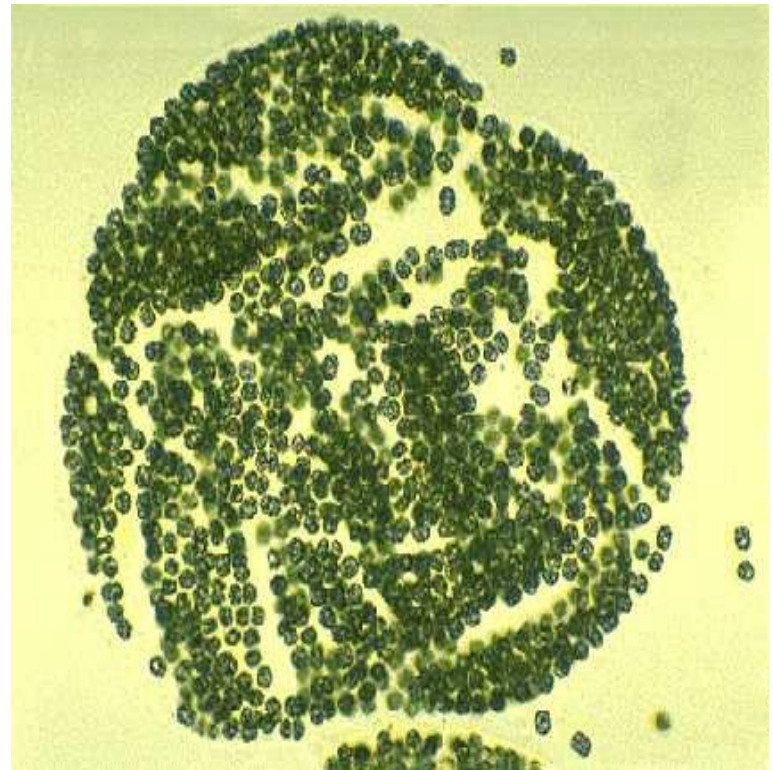
contaminated drinking water has become a major
threat.

Blue-green algae bloom



Microcystis

A toxic blue-green alga



Implications of Microbial Contamination of Water

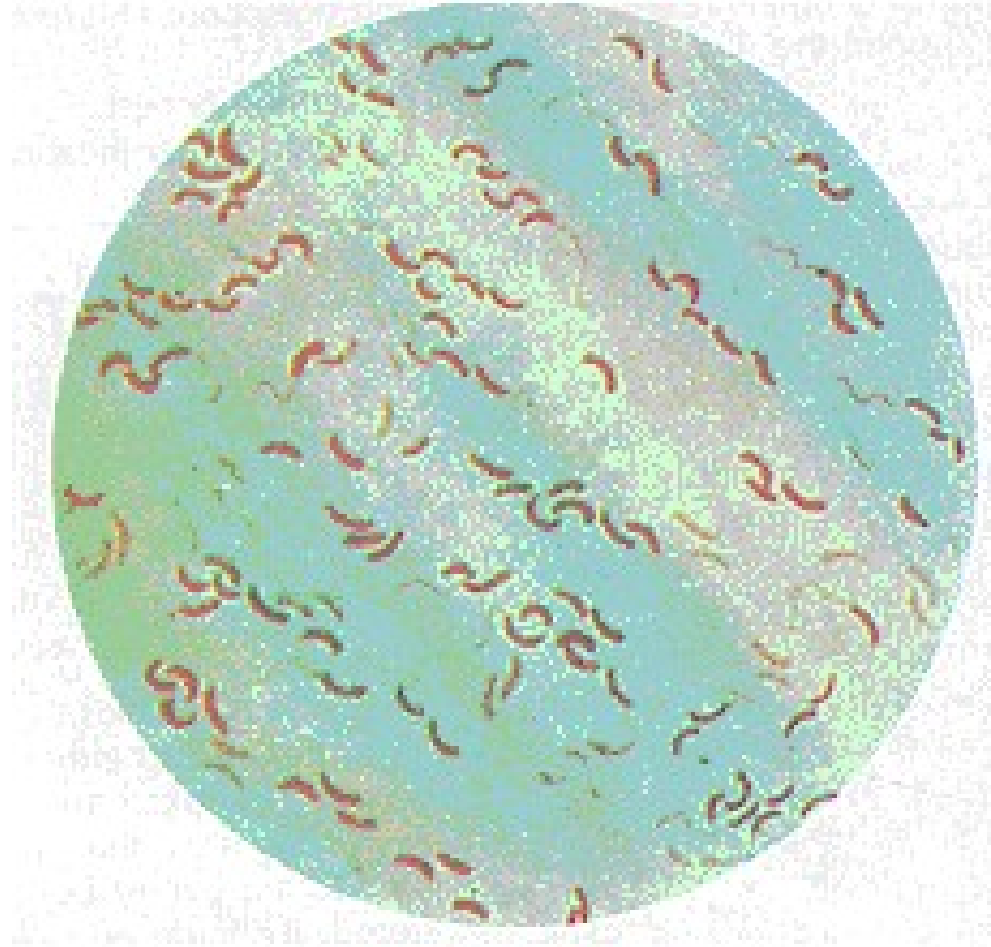
- potential for transmission of disease causing microorganisms (pathogens) via water contact - oral fecal route
- contact includes ingestion, primary contact, and secondary contact

Water Borne Diseases/Pathogens

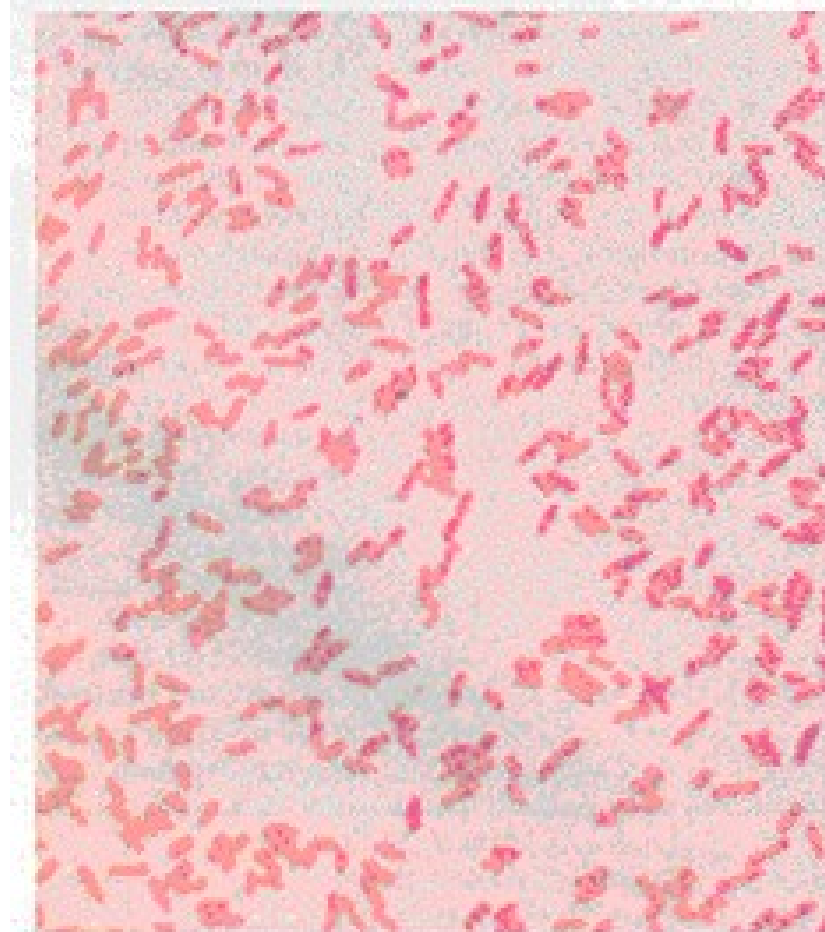
■ Bacteria

- *Vibrio cholera* - cholera
- *Yersinia enterocolitica* - gastroenteritis
- *Escherichia coli* - gastroenteritis
- *Shigella sp.*- gastroenteritis
- *Listeria monocytogenes* - flu-like symptoms
- *Salmonella sp.* - gastroenteritis, typhoid
- *Campylobacter jejuni* - gastroenteritis (animals sources)

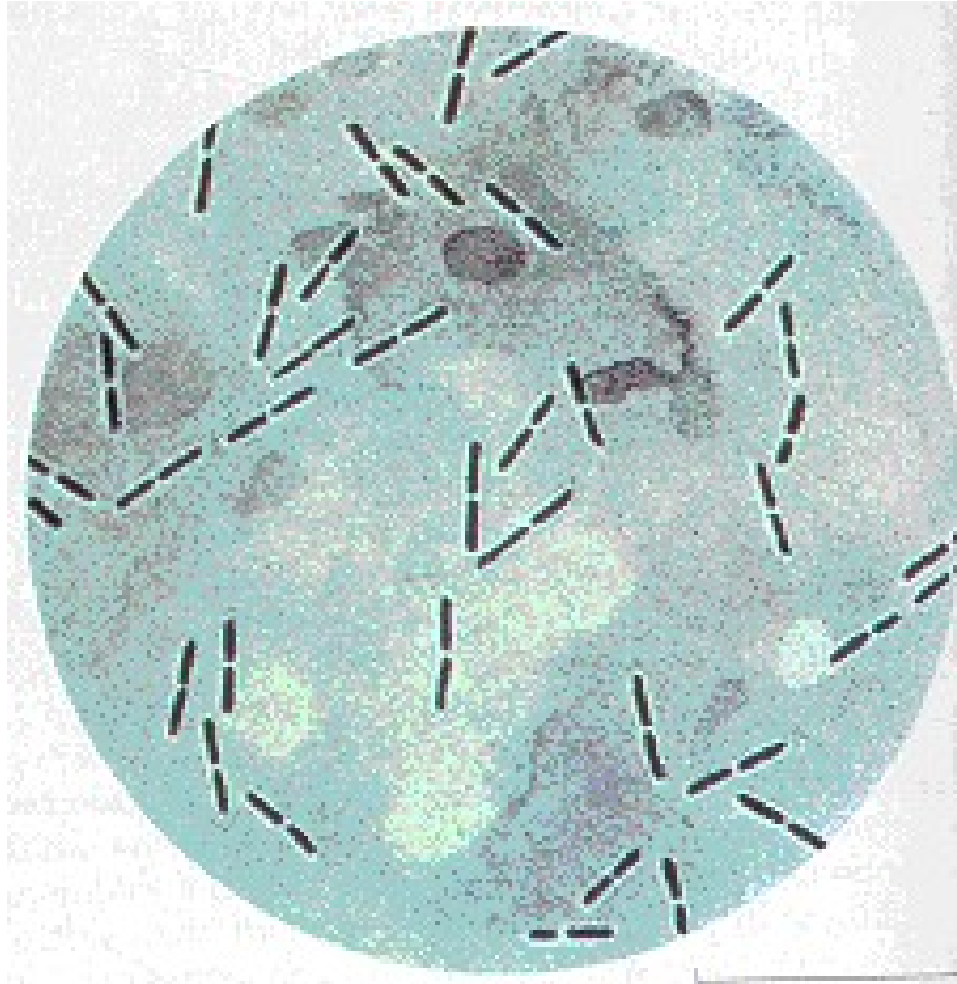
Vibrio cholerae



Yersinia enterocolitica



Clostridium ***perfringens***



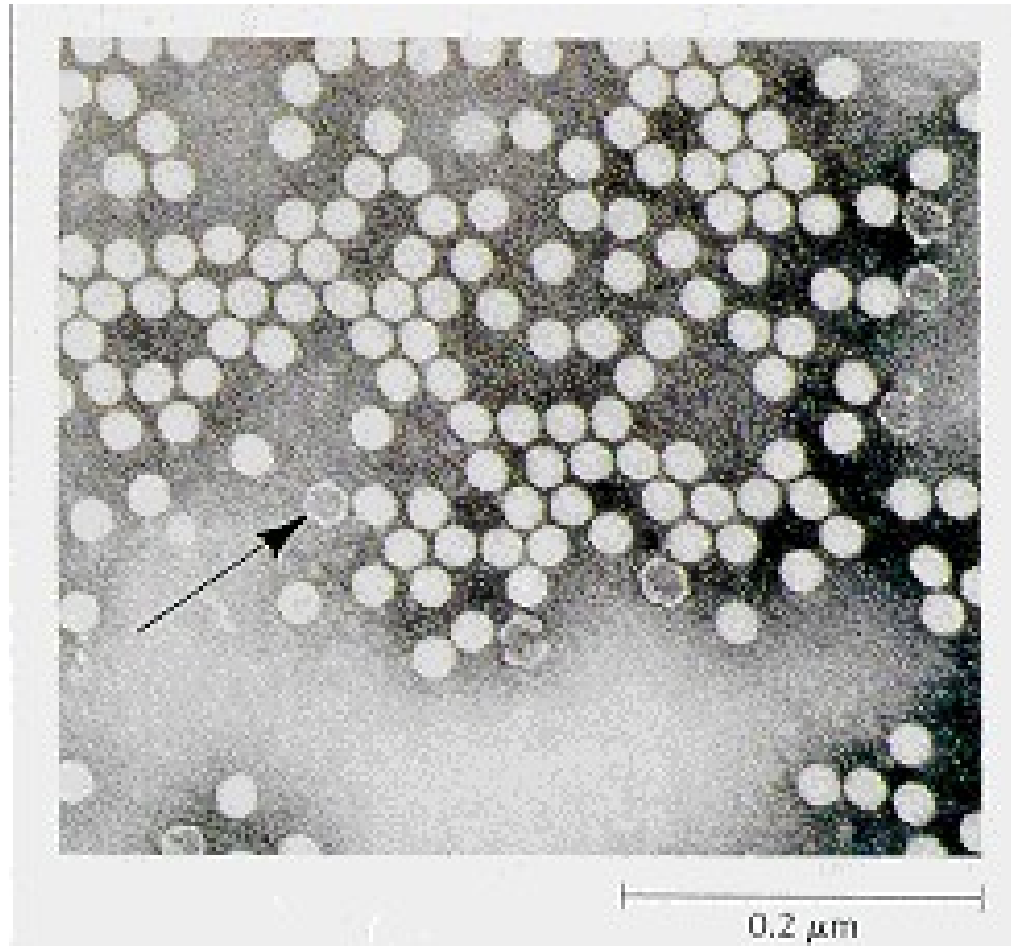
Water Borne Diseases/Pathogens

■ Viruses

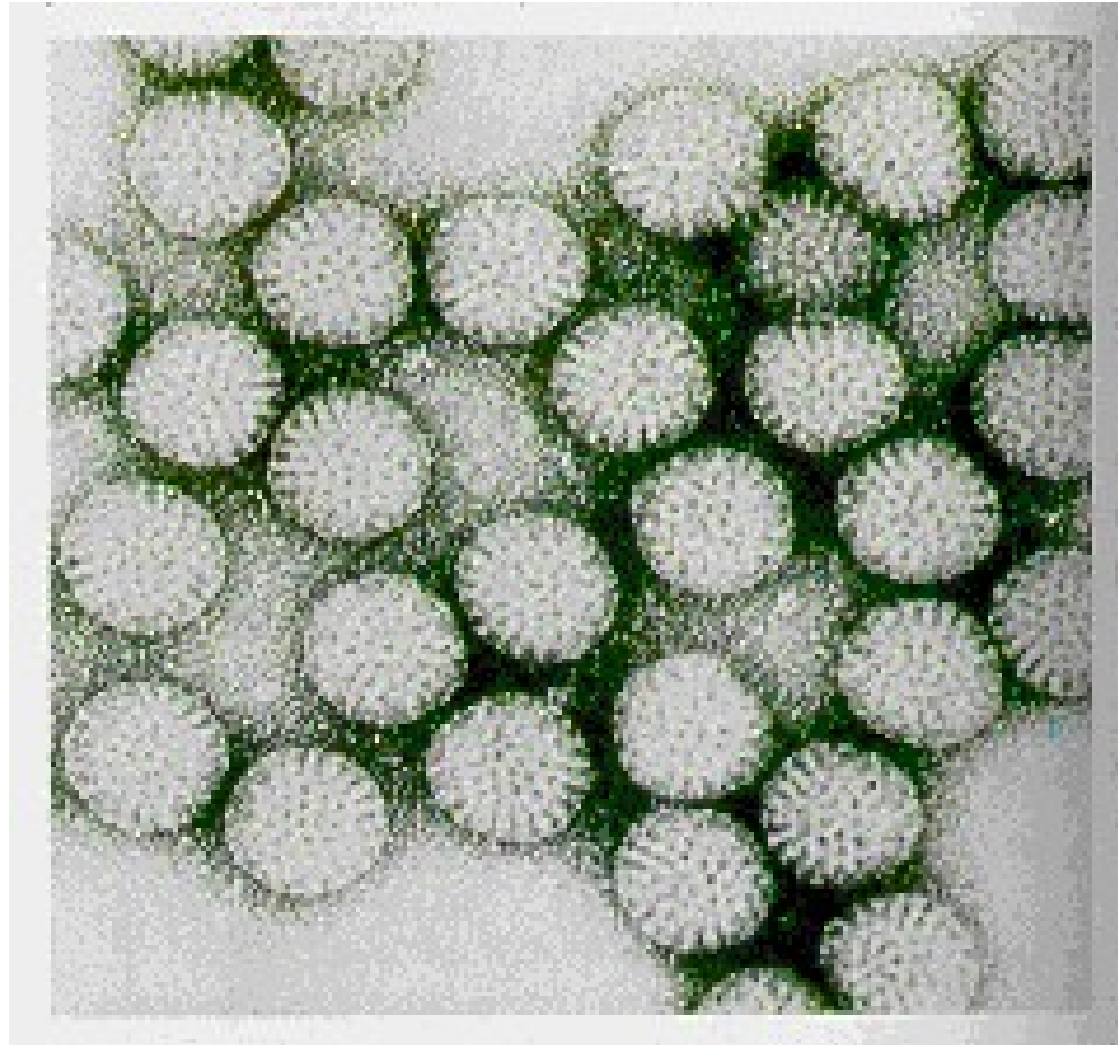
- Polio Virus - paralytic poliomyelitis
- Rotaviruses - gastroenteritis
- Norwalk Viruses - gastroenteritis
- Hepatitis Type A and E - liver inflammation

Virus

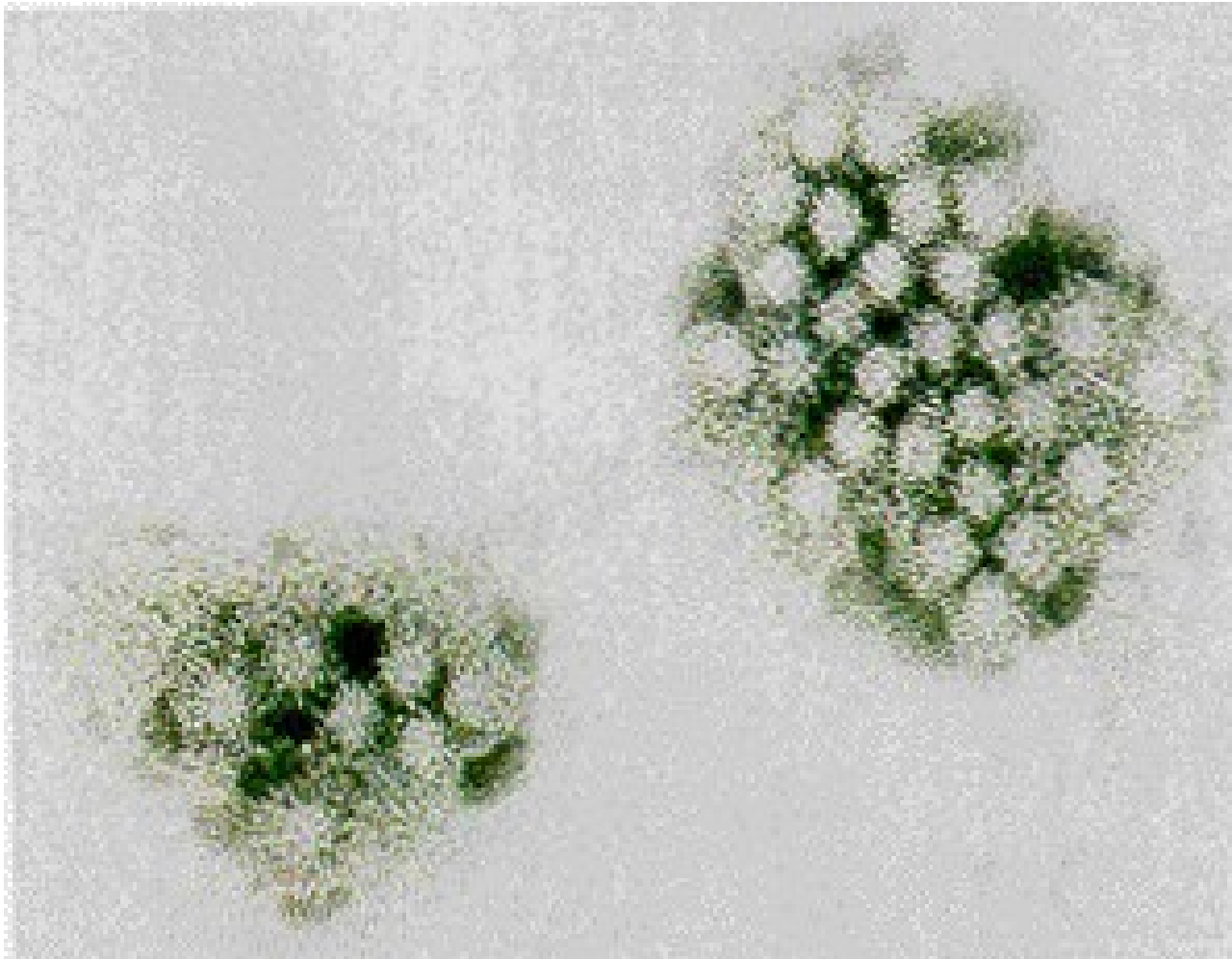
Polio



Rotavirus



Norwalk Agent



Water Borne Diseases/Pathogens

■ Protozoa

- *Entamoeba histolytica* - amoebic dysentery
- *Giardia lamblia* - intestinal disease
- *Cryptosporidium sp.* - intestinal disease

Water we drink has to be purified



Tap water



Eliminate most pathogens

purification of water

boiling

How we detect contamination?

Heterotrophic plate count (HPC)

Indicator of overall sanitation in water

HPC limits

100-500cfu/ml in tap water

50cfu/ml in bottled water (WHO, 2001).

HPC bacteria present in water samples can be determined by the pour plat method on nutrient agar, and incubated at 37C for 48 hours.

Difficulties With Routine Testing of Pathogens in Water

- present in low numbers
- limited survival time
- numerous pathogens to analyze
- time and cost prohibitive

Testing of Indicator Organisms

- need an indicator of potential pathogen contamination of water which is easy, reliable, inexpensive, quick, etc.

Requirements of Indicator Organisms

present when pathogens present in water
absent in uncontaminated water
present in higher numbers than pathogens
in contaminated water
quantity of the organism should correlate
with the amount of pollution.
survives better and longer than that of
the pathogen.
generally harmless to humans and other
animals
easy to analyze

Public water systems are required to deliver safe and reliable drinking water to their customers 24 hours a day, 365 days a year. If the water supply becomes contaminated, consumers can become seriously ill. Fortunately, many steps are being taken to ensure that the public is provided with safe drinking water. One of the most important steps is to have the water tested for coliform bacteria. Public water systems must test for coliform bacteria regularly.

What are coliform bacteria?

Coliform bacteria are organisms that are present in the environment and in the feces of all warm-blooded animals and humans.

Coliform bacteria will not likely cause illness. However, the presence of coliform bacteria in drinking water indicates that disease-causing organisms (pathogens) may be present in the water system. Most pathogens that can contaminate water supplies come from the feces of humans or animals.

Testing drinking water for all possible pathogens is complex, time-consuming, and expensive.

It is relatively easy and inexpensive to test for coliform bacteria. If coliform bacteria are found in a water sample, steps are taken to find the source of contamination and restore safe drinking water.

There are three different groups of coliform bacteria; each has a different level of risk.

Total coliform, fecal coliform, and *Escherichia coli*

Total coliform, fecal coliform, and *E. coli* are all indicators of drinking water quality. The total coliform group is a large collection of different kinds of bacteria. The fecal coliform group is a sub-group of total coliform and has fewer kinds of bacteria. *E. coli* is a sub-group of fecal coliform.

TOTAL COLIFORM, FECAL
COLIFORM AND E. COLI



When a water sample is sent to a lab, it is tested for total coliform. If total coliform is present, the sample will also be tested for *either* fecal coliform or *E. coli*, depending on the lab testing method.

Total coliform bacteria

Are commonly found in the environment (e.g., soil or vegetation) and are generally harmless. If only total coliform bacteria are detected in drinking water, the source is probably environmental. Fecal contamination is not likely.

However, if environmental contamination can enter the system, there may be a way for pathogens to enter the system.

Therefore, it is important to determine the source and to resolve the problem.

Fecal coliform bacteria

A sub-group of the total coliform group. They appear in great quantities in the intestines and feces of people and animals. The presence of fecal coliform in a drinking water sample often indicates recent fecal contamination – meaning that there is a greater risk that pathogens are present than if only total coliform bacteria is detected.

E. coli is a sub-group of the fecal coliform group. Most *E. coli* are harmless and are found in great quantities in the intestines of people and warm-blooded animals. Some strains, however, may cause illness.

The presence of *E. coli* in a drinking water sample almost always indicates recent fecal contamination – meaning that there is a greater risk that pathogens are present.

WHO bacteriological quality of drinking water (WHO, 1996)

- *E.coli* or thermotolerant coliform bacteria -Must not be detectable in any 100ml sample
- Total coliform bacteria -Must not be detectable in any 100ml sample

A Note about *E. coli*:

Most outbreaks have been related to food contamination, caused by a specific strain of *E. coli* known as *E. coli O157:H7*. When a drinking water sample is reported as “*E. coli* present” it does not mean that this specific strain is present and in fact, it is probably not present. However, it does indicate recent fecal contamination.

Boiling, or treating contaminated drinking water with a disinfectant, destroys all forms of *E. coli*, including *O157:H7*.

Coliform Bacteria - Indicators of Fecal Contamination

bacteria in the family

Enterobacteriaceae

generally include the genera

Enterobacter, *Klebsiella*, *Citrobacter*,
and *Escherichia* and *Aeromonas* sp.

present in the intestinal tract of

numerous organisms

gram negative, non-spore forming
rods

ferment lactose with acid and gas
production in 48 hours at 35 °C

Family - ENTEROBACTERIACEAE

Coliforms

1. Total coliforms- *Escherichia* sp.
Enterobacter sp., *Citrobacter* sp.,
Klebsilla sp., *Serratia* sp., *Yersinia*
sp.
2. Fecal coliforms – *Escherichia coli*,
3. Non Coliforms – *Salmonella* sp.
Shigella sp. *Proteus* sp. *Erwinia* sp.
Hafnia sp.